Jordan Hill

Machine learning research · Signal-processing architectures · Research engineering
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Profile

Machine learning researcher developing **Kairos**, a novel neural architecture in which an incoming signal, its learned representations and a language model advance at different rates while remaining responsive to one another. Its alignment premise: AI systems that share our relation to time, meeting each moment from within an unfolding situation as we do, should be less alien to us, and therefore more predictable. Builds and runs the full research stack without an infrastructure team, from gradient-checked reference code to an agent-native experiment system. PhD candidate in Artificial Intelligence at UWA, supervised across philosophy, computer science and linguistics, bringing conceptual depth on timing, language and alignment to technical work.

Research

Kairos — a novel neural architecture for signal processing | 2026–present
Prosody Labs and doctoral research, University of Western Australia; unpublished.

- **Architecture.** Learned encoders and decoders let a sequence-model core (Gated DeltaNet, attention or hybrid) run at its own rate, independent of the rates at which signals enter and leave, so one architecture serves raw audio, text and both together.
- **Proof-of-concept audio reconstruction.** In the Kairos audio playground, passes raw 8 kHz audio through learned representations and a fixed-rate 1 kHz bottleneck (8× fewer positions), reconstructing it at about **23 dB SNR** on a recorded clip.
- **Full-duplex conversation.** Extends Moshi (Défossez et al., 2024), the basis of NVIDIA's PersonaPlex, which hears and speaks through a separately trained codec quantising 24 kHz speech into eight discrete codebooks at a fixed 12.5 Hz: Kairos instead learns continuous representations from raw waveform end to end, on audio and text clocks that advance independently, scaling the playground's reconstruction to the full English LibriVox corpus. Trains 0.8-billion-parameter models from Qwen3.5-0.8B on four NVIDIA H100s, against a Moshi-recipe replication at the same scale.
- **Byte-level language modelling from signals.** Fuses three ideas into one model: text encoded as a signal that Kairos processes like audio; byte prediction, whose 256-symbol vocabulary keeps prediction heads cheap where token vocabularies make them a memory bottleneck; and multi-token prediction extended from 8 bytes ahead (Gloeckle et al., 2024) to 32, scaling to 64, so a tokenizer-free model is trained to look past characters to word-scale structure. Trained on four AMD MI210s and NVIDIA H100s, evaluated on the CORE benchmark suite, alongside an original recurrent token mixer.
- **Evidence-led redesign.** Earlier versions in 120–150M-parameter language models redirected it to variable-rate processing.

Research engineering

- At Prosody Labs, built **Execute**, the experiment system behind Kairos: every run is a commit-pinned, auditable record, queued via git by hand or by AI coding agents on two GPU clusters with guardrails for unattended use; 1,100+ runs since July 2026.
- Maintains a gradient-checked reference model and a PyTorch trainer validated against it, so results rest on verified maths.
- Writes custom **Triton kernels for both NVIDIA CUDA and AMD ROCm**, qualified on RTX 4090 and MI210, and trains with FSDP/DDP across H100 and MI210 clusters; **2.26× faster training updates** in a matched four-H100 benchmark.
- Operates a self-hosted k3s platform with TLS and single sign-on, serving live training playgrounds, run review and analysis.
- Directs AI coding agents against explicit acceptance criteria and numerical checks, recording failures alongside successes.

Presentation

"Temporal positionality in conversational AI: why timing is not latency." Accepted talk, Australian Linguistic Society Conference (ALS 2026), UWA, December 2026.

Technical skills

ML: PyTorch; recurrent, state-space and attention models; audio and language modelling; FSDP/DDP on CUDA and ROCm.
Engineering: Python, Triton, JavaScript, Slurm, Kubernetes, Docker, Git, FastAPI; experimental design and technical writing.

Experience

Founder and Director, Prosody Labs, Perth | 2024–present. Founded the AI research company developing Kairos and Execute.

Lecturer, AI and Data Science, North Metropolitan TAFE | August 2023–present. Built and leads the AI programme, including the ICTSS00120 AI Skill Set and Certificate IV (Data Science and AI) pathway.

AI Specialist Technical Committee, Future Skills Organisation | current. Advises the national ICT training package review on AI qualifications as a subject-matter expert and educator, for the Australian Government's Jobs and Skills Council.

Machine Learning Engineer / Research Assistant, AIMES Laboratory, University of Melbourne | 2022. Built deployment automation, Kubernetes infrastructure, data pipelines and training workflows for an applied ML project.

Software engineering, InSig Technologies and Risk Management Technologies | 2020–2021. Developed production systems: a mining information system (Solutions Architect) and Java/AngularJS risk-management software.

Education

PhD in Artificial Intelligence, University of Western Australia | 2026–present. Thesis: *Kairos: A Series of Unresolved Analogies Between Two Natures*; expected submission August 2029; RTP Scholarship. Supervisors: A/Prof Nin Kirkham (Philosophy), Dr Wei Liu (Computer Science), A/Prof Celeste Rodriguez Louro (Linguistics).

Graduate Certificate in Data Science, RMIT University | 2024. GPA 3.8.

Master of Arts in the Liberal Arts, St John's College, Annapolis | 2023. GPA 3.95 in Ancient Greek; Ramsay Scholarship.

Bachelor of Arts (Honours), First Class, History, University of Western Australia | 2019. Philippa Maddern Memorial Prize.